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ECE 49595II

August 29, 2026

Job Matching Algorithm Specification

AI Disclosure: AI was not used in the creation of this document.

Introduction

From the CDR feedback, it has become clear that our job matching algorithm is not well-defined. This issue has a major impact on the quality of our product given that, as of the current spec, the main benefits we offer over platforms like LinkedIn is the supposed accuracy of our job search feature. The improvements in accuracy we have over our competitors is vital to our ability to stand out in this already-crowded market, so we should know our implementation plan at an early stage.

Upon reviewing team notes, my individual assignments, and my own understanding of our implementation, it became clear to me that part of this was an issue of inter-team communication, but also of treating this feature as a black box due to the use of LLMs. As such, this document will be reviewed by all team members to ensure we are on the same page with regards to the job search feature. To begin with, I will present a matching algorithm that assumes the LLM is a perfectly-accurate oracle, then describe what modifications will need to be made in response to possible biases the system actually ends up having.

Oracle Implementation

We will assume that the oracle has generated a database of all possible skills jobs may require, the skills that each applicant holds, and the skills required by each job. Details such as salary and location will also be included in the listing database, and each user will have their preferences for each recorded. These entries will also be marked with a number from 1 to 4 describing how proficient an applicant is in a given skill or how important that skill is to a given job. The details for the creation of these databases is described in my Design Alternatives individual assignment but is not currently relevant to the scope of this design.

Compatibility score

The first step of job matching is to calculate a compatibility score only considering the skills required by the job and demonstrated by the applicant. The formula for the compatibility score is displayed in figure 1, where A is the applicant, L is the job listing, C_{AL} is the compatibility score of applicant A with listing L, S_L is the list of skills required by listing L, and P_{sX} is the proficiency level required/demonstrated by applicant/listing X for skill s.

$$C_{AL} = \sum_{i}^{S_L} \frac{P_{iL}}{4} * \frac{P_{iA}}{4}$$

Figure 1: compatibility score calculation

For the sake of proper weighting with other factors, the compatibility score should be normalized such that the maximum possible compatibility score for any candidate on this listing is 1.0.

Location Filtering

The second factor of job matching is location filtering, typically set in the search parameters. This is exclusively used to filter out jobs outside of the radius and is not included in the ordering of listings due to usage patterns. If a user expands their search radius beyond a currently-small default, it is because they want to see jobs further away from them, so it is not helpful to penalize jobs based on distance. Similarly, a user can constrain the search radius if they are receiving recommendations too far away from them.

Other factors may also be used to filter out irrelevant jobs, but for much the same reasons as location, they will not be included in

Salary

The third factor used in job searching is desired salary. When calculating the search index for a listing, this factor is included by taking the ratio of the offered salary to the user's desired salary, then applying a ceiling of 2.0. If a listing does not disclose payment details, then this should default to 0.25.

Final Search Index

The search index is calculated via a weighted sum of compatibility score with the salary score. Given that job compatibility is generally more important to applicants than salary, the initial weighting should be 0.7 for compatibility score and 0.3 for salary score, especially since the salary score can go up to 2.0. Finally, job listings are returned to the user sorted in decreasing order by their search index.

Potential Modifications

Since we know that LLMs often are not oracles, here are some changes that may need to be made to improve the accuracy of jobs provided to the user.

Proficiency Level Granularity

From my limited testing, it appears that ChatGPT will generate a large number of skills for a job that may be only minorly relevant, so the fact that missing four minimally-relevant skills are equivalent to missing 1 extremely important skill may harm job search accuracy. As such, we may need to increase the maximum proficiency level, potentially to using a percentile system, but more likely to 8 or 16.

Prompt Engineering

When not given direction, ChatGPT seems to be biased towards STEM jobs and skillsets in particular, so if situations like this occur, some extra direction might need to be included inside of the prompt used to generate the skills database. I would first recommend using [Directional Stimulus Prompting] to increase accuracy, but the team may decide to add additional guardrails in natural language if required.

Weighted Sum Modification

If it appears that salary or compatibility are prioritized too heavily with the current setup, the weighted sum may need to be biased in the direction of either. In addition, if it is found that a large percentage of listings don't include salary information, it may be beneficial to change the default salary score when none is listed to a higher value

Geometric Skill Proficiency Prioritization

As an additional guardrail for a large number of low-priority skills masking for a severe lack in high-priority skills when matching candidates, the P_{iL} term in compatibility score calculation can be squared or otherwise given a higher degree than 1 to reduce the impact low-priority skills have in job searching.

Specific Definitions for Proficiency Levels

If the LLM struggles to give consistent or accurate proficiency levels to candidates or listings, the prompt should be modified to include specific definitions for what each proficiency level entails. For example, proficiency 1 could mean 1+ years of non-professional experience, proficiency 2 could mean 1+ year of professional/academic experience, proficiency 3 could mean 3+ years of professional/academic experience, and proficiency 4 could mean 5+ years of professional experience.